6-7 Digitalization of Sound

Point!

1 Sound

(1) Sound : A phenomenon transmitted through the vibration of air. Waveform ([1]**analog**) data.
- For audio data on CDs, etc., sound is converted into ([2]**digital**) data.

(2) ([3]**Frequency**) : The number of waves contained in one second.
The unit is ([4]**hertz**) [Symbol: ([5]**Hz**)].

(3) ([6]**Period**) : The time it takes for one wave to propagate. The unit is ([7]**seconds**).

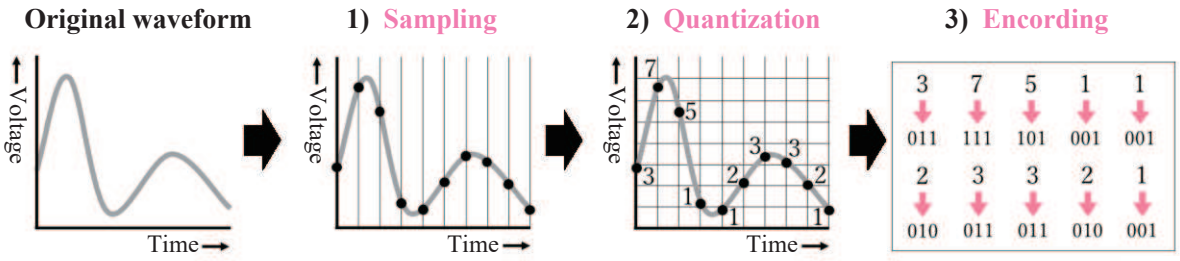


2 Digitalization of Sound

(1) ([8]**Pulse Code Modulation (PCM) method**) : A method for digitizing audio analog data.
Convert sound information into binary code.

<Steps for digitizing sound using the PCM method>

1) ([9]**Sampling**) : The horizontal axis (time) is divided at regular intervals, and the wave height (voltage strength) is extracted. The extracted points are referred to as ([10]**sample points**).
   - ([11]**Sampling period**) : The time interval used to divide when sampling.
   - ([12]**Sampling frequency**) : The number of samples taken per second.
2) ([13]**Quantization**) : The vertical axis (voltage) is divided at regular intervals, and the wave heights obtained from sampling are converted to the nearest values on the vertical scale.
   - ([14]**Quantization bit depth**) : This determines the number of levels into which the range is divided during quantization.
3) ([15]**Encoding**) : Express the quantized values in binary form.



(2) Digitization and Data Volume
As the sampling frequency and the quantization bit depth ([16]**increase**), the sound becomes closer to the original analog waveform. This causes the sound quality to ([17]**improve**); however, the amount of data also ([18]**increases**).

(3) ([19]**Sampling theorem**) : If the sampling frequency exceeds ([20]**twice**) the highest frequency contained in the original analog waveform, the waveform of the original analog signal can be accurately reconstructed from the digitized data.

3 Amount of Sound Data

(1) Channels : The number of signals used when transmitting sound. A method of playback using a single signal is called ([21]**monaural**), and a method of playback using two different signals is called ([22]**stereo**).

(2) How to Calculate the Amount of Audio Data
Data amount [bits] per second = ([23]**Sampling frequency × Quantization bit depth × Number of channels**)

Warm Up

Answer the following questions.

(1) Choose the one appropriate procedure for digitizing an analog sound signal from the options **A** to **D**.

A Sampling → Quantization → Encoding
B Encoding → Quantization → Sampling
C Quantization → Sampling → Encoding
D Quantization → Encoding → Sampling

(2) Calculate the data amount for 1 second in kilobytes (KB) when digitizing music at a sampling frequency of 44,100 Hz, a quantization bit depth of 16 bits, and in 2-channel stereo. However, assume that 1 KB = 1,000 B, and round off to the nearest whole number.

(3) Calculate the data amount for 1 minute in megabytes (MB) when digitizing music at a sampling frequency of 44.1 kHz, a quantization bit depth of 24 bits, and in 2-channel stereo. However, assume that 1 KB = 1,000 B, and 1 MB = 1,000 KB, and round off to the nearest whole number.

Explanation

(1) **A**

(2) Data amount [bits] per second = Sampling frequency × Quantization bit depth × Number of channels
Therefore, the amount of data per second is 44,100 [Hz] × 16 [bits] × 2 [channels] = 1,411,200 [bits].
Since 1 B = 8 bits, 1,411,200 [bits] ÷ 8 = 176,400 [B]
1 KB = 1,000 B, so 176,400 [B] ÷ 1,000 = 176.4 ≈ 176 [KB] 176 KB

(3) 44.1 kHz = 44.1 × 1,000 = 44,100 Hz, and since 2 minutes = 120 seconds, the data amount per minute is 44,100 [Hz] × 24 [bits] × 2 [channels] × 60 [seconds] = 127,008,000 [bits]
Since 1 B = 8 bits, 127,008,000 [bits] ÷ 8 = 15,876,000 [B].
1 KB = 1,000 B, 1 GB = 1,000 MB, therefore,
15,876,000 [B] ÷ 1,000 ÷ 1,000 = 15.876 [MB] ≈ 16 [MB] 16 MB

Try

1 Answer the following questions.

(1) Choose the term that best fits into the blanks [1] to [10] in the following sentences from the options **A** to **M** below, and answer using the letters.

> Sound is a phenomenon that is transmitted through the vibrations of air. The number of waves contained in one second is called ([1]) and is expressed in units of ([2]). The time it takes for one wave to propagate is called the ([3]), and it is expressed in seconds. These vibrations are ([4]) data. In contrast, sound information stored on DVDs, CDs, etc. is ([5]) data. Next, follow these steps to convert the data from ([4]) to ([5]).
>
> First, the process of dividing sound waves at regular time intervals and extracting the amplitude values (voltage) for each interval is called ([6]). Next, divide the vertical axis (voltage) at regular intervals and determine the step values. This is called ([7]), and the component that determines the quantization levels is called ([8]). Typically, with n bits, there are 2^n levels. Finally, ([9]) refers to the representation of values obtained through ([7]) in binary form. The method of converting information such as sound into binary codes by replacing numerical values in binary with two different voltage levels (high and low) is called the ([10]) method.

A	Encoding	**B**	Sampling	**C**	Quantization	**D**	Frequency	**E**	Period
F	Analog	**G**	Digital	**H**	Sampling period	**I**	Quantization bit depth		
J	PCM	**K**	MIDI	**L**	Hertz	**M**	Second		

(2) Choose the one that enables waveform restoration closest to the original analog signal when digitizing an analog audio signal from the options **A** to **D**.

A Increase the sampling frequency and increase the quantization bit depth.
B Increase the sampling frequency and decrease the quantization bit depth.
C Decrease the sampling frequency and increase the quantization bit depth.
D Decrease the sampling frequency and decrease the quantization bit depth.

2 Answer the following questions. In these questions, assume that 1 KB = 1,000 B, and 1 MB = 1,000 KB.

(1) What is the data amount for 1 second in kilobytes (KB) when digitizing music at a sampling frequency of 44,100 Hz, a quantization bit depth of 16 bits, and in 1-channel monaural? Round off to the nearest whole number in your answer.

(2) What is the data amount for 1 minute in KB when digitizing music at a sampling frequency of 192 kHz, a quantization bit depth of 24 bits, and in 2-channel stereo?

(3) What is the data amount for 3 minutes in MB when digitizing music at a sampling frequency of 44.1 kHz, a quantization bit depth of 16 bits, and in 2-channel stereo? If the length of one song is 3 minutes, approximately how many songs can be stored on a 650 MB CD? Round off to the nearest whole number in your answer.

Exercise

1 Answer the following questions.

(1) Choose the term that best fits into the blanks [1] to [6] in the following sentences from the options **A** to **I** below, and answer using the letters.

> Sound is analog data in waveforms that are transmitted through the vibration of air. The process of capturing sound waveforms on a computer begins with segmenting the waveforms at regular time intervals. This process is called ([1]). The number of partitions per second is called the ([1]) frequency. Next, convert and approximate the segmented wave heights into a continuous bar graph. This process is called ([2]). At this time, the value indicating how many levels the numbers are expressed in is called ([3]). Finally, perform ([4]), which uses 0s and 1s to express the value in binary after ([2]). This method of converting sound into digital data is called ([5]).
>
> In the case of digitizing sound, digital data converted from analog data will approximate the original waveform more closely if the ([1]) frequency is higher. Furthermore, the greater the ([3]), the more accurately the waveform can be represented numerically. However, as a result, the amount of data becomes ([6]).

A Encoding	**B** Sampling	**C** Quantization	**D** Larger	**E** Smaller
F Sampling period	**G** Quantization bit depth	**H** PCM method	**I** MIDI format	

(2) The following statements **A** to **D** are about the sampling frequency and the number of quantization levels. Mark "○" if the statement is correct, and "×" if it is incorrect.

A Sampling frequency refers to the number of times sampling occurs in one second.

B The greater the number of levels during quantization, the larger the amount of data becomes.

C If you want to approximate the original analog waveform more closely, it is effective to increase the sampling frequency and reduce the number of levels during quantization.

D If the maximum frequency of the original analog waveform is 40 Hz, you should digitize it with a sampling frequency greater than 80 Hz.

2 Answer the following questions. In these questions, assume that 1 KB = 1,000 B, and 1 MB = 1,000 KB.

(1) What is the data amount for 10 seconds in kilobytes (KB) when digitizing music at a sampling frequency of 44,100 Hz, a quantization bit depth of 16 bits, and in 1-channel monaural?

(2) What is the data amount for 1 minute in MB when digitizing music at a sampling frequency of 96 kHz, a quantization bit depth of 24 bits, and in 2-channel stereo? Round off to the nearest whole number in your answer.